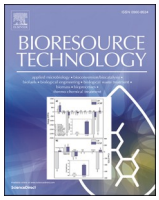
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Effects of biological pretreatments of microalgae on hydrolysis, biomethane potential and microbial community

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HIGHLIGHTS

- Hydrolysis and BMP performance of various biological pretreatments were studied.
- Different pretreatment conditions have resulted in different microbial diversity.
- PCA analysis identified no correlation between hydrolysis and BMP.
- BMP showed good correlation with sN, sP, sCOD, and propionic acid formation.
- Total VFAs were correlated to VS removal and levels of acetic and butyric acid.

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ABSTRACT

Parameters of temperature-phased anaerobic digestion (TPAD) were varied to study their effects on hydrolysis, biomethane potential (BMP), and microbial diversity of microalgae biodegradation. Anaerobic pretreatments at 85 °C demonstrated the release of soluble carbohydrate and protein molecules under low microbial metabolic activity. However, at 55 °C, anaerobic pretreatments showed superior performance in methane yield, nutrient release, and volatile fatty acids (VFAs) production due to dominant *Clostridium.* Furthermore, the highest destruction of volatile solids (VS) was observed during aerobic pretreatments at 55 °C under the influence of various quantities of these genera – *Luteimonas*, *Symbiobacterium, Soehngenia, Thermobacillus,* and *Ureibacillus*. Statistical analysis revealed that hydrolysis and BMP were not correlated. However, soluble nitrogen and phosphorous showed strong correlation with methane (r = 0.623 and 0.948, respectively) under thermo-anaerobic pretreatment, while VS removal and concentrations of acetic and butyric acids and lipids were positively correlated with each other under thermo-aerobic pretreatment.

1. Introduction

As a potential alternative energy source to fossil fuels, third-generation biofuels derived from microalgae have shown promise for bioenergy production due to their high quantum yield, high biomass per unit area, and low competition for arable land (Marsolek et al., 2014). Growing algae using wastewater effluent offers multiple benefits such as the recovery of nutrients, removal of heavy metals, and production of biogas as a renewable energy source. It also has the potential of coupling CO_2 mitigation to wastewater recycling (Barragán-Trinidad et al., 2017).

However, digestion of algae requires pretreatment to breakdown resistant cell walls to render them susceptible to microbial attack (Gonzalez-Fernandez et al., 2015). Various technologies have been suggested as pretreatment including physical, thermal, chemical, mechanical, and biological processes (Passos et al., 2014). Biological degradation of microalgae using anaerobic digestion offers the most cost-effective, environmental-friendly option, is feasible for large-scale

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application, and is energy-efficient, requiring few chemicals, while producing non-inhibitory byproducts (Sindhu et al., 2016). It requires microorganisms such as fungi, protozoa, and bacteria, and costly industrial enzymatic cocktails (Barragán-Trinidad et al., 2017).

Anaerobic digestion (AD) is generally regarded as a potential means to recovery energy via biogas production. Among the various types of AD setups, the current study utilized temperature-phased anaerobic digestion (TPAD) due to its excellent reactor stability, methanogen activity, and the abundance and diversity of the microbial community (Ge et al., 2011; Jang et al., 2014). The biggest advantage of TPAD is its potential to combine both enhanced hydrolysis and stability while utilizing moderate energy input and completely removing pathogens (Pervin et al., 2013). Moreover, TPAD provides both high destruction of solids along with an increased rate of gas production at a relatively low capital cost (Ge et al., 2011). This increased removal of volatile solids (VS) thereby reduces the costs associated with subsequent sludge dewatering and sludge disposal (Lv et al., 2010).

However, applications of these biological methods to microalgae pretreatment have been limited, to date, due to insufficient knowledge and uncertainties about the process performance. Among these, the literature reveals insufficient studies to demonstrate hydrolysis, VFAs formation, and BMP capabilities of different TPAD systems during microalgae digestion. Moreover, understanding the correlations between these factors helps to optimize as well as indirectly estimate BMP and VFAs which would otherwise require us to employ tedious, expensive, and time-consuming laboratory procedures. Moreover, much remains unknown about the relationship between the dynamics of microbial diversity and biological pretreatments while varying hydraulic retention time (HRT), temperature, and redox conditions (Lv et al., 2010). The scientific literature contains several related studies that applied statistical approaches. Perendeci et al. (2019) considered various substrate compositions including glucose, Kjeldahl nitrogen, and chlorophyll in the mesophilic co-digestion of microalgae to create statistical relationships with BMP. Lesteur et al. (2010) correlated BMP of mesophilic biodegradation with elemental and biochemical composition. Magdalena et al. (2019) employed organic loading rate and ammonia concentration to observe the production rate of VFAs from microalgae digestion under mesophilic conditions. Very few studies have compared performance and application of different thermophilic pretreatment phases of TPAD for microalgae digestion. Zamorano-López et al. (2019) studied microbial dynamics and its statistical correlation with organic loading rate and free ammonia in a TPAD system. Some studies addressed the effect of pretreatment temperature in TPAD (Ge et al., 2011) while others compared the BMP potential of TPAD and mesophilic conditions (Zhang et al., 2019). However, to the best of our knowledge, there has been no comprehensive study comparing the pretreatment performance of different TPAD-based biological pretreatments of microalgae, and no study has clearly identified indicators to estimate the extent of hydrolysis, tendency of VFAs formation, and BMP through considering their interactions with the underlying microbial community.

The main objective of this study was, therefore, to investigate the performance of different TPAD systems under variable temperatures, HRTs, and redox conditions and statistically compare their response for identification of parameters sufficiently correlated to forecast and optimize hydrolysis propensity, VFAs production and BMP estimation of microalgae biodegradation. This study uniquely provides extensive information on the microbiomes of different thermophilic pretreatment systems, substrate physicochemical properties, and biodegradation responses all in one factorial plane allowing a comprehensive overview of their correlation, interaction, and comparison with different biological pretreatments of microalgae including the thermo-aerobic process which has not previously been dealt with in detail.

2. Materials and methods

2.1. Microalgal biomass and the inoculum

The *Chlorella vulgaris* microalgae biomass utilized as a substrate in this study was cultivated in freshwater and purchased from Daesang Corporation, Korea. The biomass was grown in complete darkness at a temperature of 30 °C and aerated using air spargers at a flow rate of 1 L/min while being continuously agitated at a speed of 90 rpm. To facilitate production, microalgae growth was assisted using 8–10% liquid sugar. The pH of the algae growth media was maintained between 7.5 and 8.0. During pretreatment, a semi-continuous mode setup was established and tap water had been utilized to dilute the microalgae samples due to their high concentration of total solids (TS). The inoculum for microalgae pretreatment was acquired from a full-scale mesophilic anaerobic digester at the Gwangju municipal wastewater treatment plant (WWTP) (Gwangju, Korea). For pretreatment under aerobic conditions, the inoculum was obtained from activated sludge in the secondary treatment of the same plant. Detailed characteristics of the raw microalgae and inocula are shown in Table 1.

2.2. Microalgae pretreatment (first phase of TPAD)

Raw microalgae were diluted twice and the pH was set to 7.0 before starting the experiment. Then, the microalgae as a substrate were semi-continuously fed into the reactor in a thermophilic, continuously stirred 5 L tank reactor having a working volume of 2.8 L and was operated for 908 h. This duration refers to the time consumed for the entire operation beginning with the start-up and adaptation phase (0 to 300 h), then pretreatment at an HRT of 3 days (300 to 624 h), pretreatment at an HRT of 2 days (624 to 792 h), and pretreatment at an HRT of 1 day (792 to 908 h). See the time boundaries in Fig. 1. The continuous agitation stirring speed was 150 rpm. To maintain aerobic conditions, an initial flow rate of 5 L/min was achieved by the air spargers to aerate the reactor, then was continuously increased until bubbles were formed at a flow rate of 10 L/min. For establishing anaerobic conditions, the reactor was purged once with nitrogen at the beginning of the experiment. To change the temperature in the reactor, heating tape was used. The experiment was started in such a way that, originally, the anaerobic conditions were set at 85 °C, 70 °C, and 55 °C. The assay from 70 °C was ultimately omitted from the checklist since there was no significant difference between its results and those from 85 °C and 70 °C. Anaerobic conditions at 85 °C and 55 °C were finally selected and run at 3 different hydraulic retention times (HRTs) (3 days, 2 days, and 1 day). Then, the operation was carried out under aerobic conditions at 55 °C using the same HRTs (3 days, 2 days, and 1 day). A syringe was used to take digested microalgae samples 2 to 3 times a day with sample volumes of 450 to 900 mL depending on the HRTs. All data were collected either in duplicate or triplicate and the average and standard error values were recorded. After completion of the microalgae pretreatment, BMP was

Table 1
Characteristics of the undiluted raw microalgae biomass and the inocula.

Parameter	Microalgae*	Inocula
pH	7.7	7.61
Total solids, TS (g/L)	114.3 ± 0.9	27.77 ± 0.8
Volatile solids, VS (g/L)	108.3 ± 0.2	17.30 ± 2.1
Total chemical oxygen demand, tCOD (g/L)	103.5 ± 0.2	14.31 ± 0.3
Soluble chemical oxygen demand, sCOD (g/L)	5.4 ± 0.0	4.25 ± 0.1
Soluble carbohydrate (g/L)	4.2 ± 0.1	–
Soluble protein (g/L)	1.1 ± 0.1	2.66 ± 0.4
Total nitrogen, TN (g/L)	10.6 ± 0.1	–
Soluble total nitrogen, sN (g/L)	0.48 ± 0.08	–
Total phosphorous, TP (g/L)	2.5 ± 0.8	–
Soluble total phosphate, sP(mg/L)	75.5 ± 0.5	–
Lipids	0.7 ± 0.21	–

*values provided in Table 1 should first be halved.

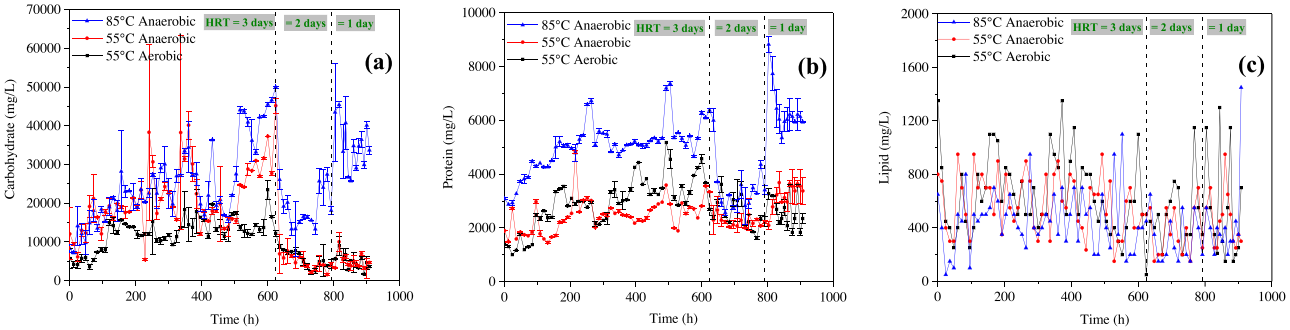


Fig. 1. Amounts of macromolecules released during different biological pretreatments at different HRTs: (a) carbohydrates; (b) proteins; (c) lipids.

tested for each regiment based on the redox condition, temperature, and HRT.

2.3. Biochemical methane potential assay (Second phase of TPAD)

The AD of untreated microalgae was set as a reference to measure methane yield. Both raw and pretreated microalgae were tested under 12 different pretreatment conditions in a semi-continuous mode. During the biochemical methane potential experiment, the biologically pretreated microalgae were digested in a 250 mL serum bottle with a working volume of 150 mL. The ratio of VS-based inoculum to pretreated microalgae in the reactor was 2:1 and the initial pH was 7. Mesophilic conditions were established by adjusting the temperature to 35 °C and flushing the headspace with nitrogen gas. The test bottles were completely plugged with stoppers and placed on a shaking incubator running at 150 rpm. The BMP tests continued until the methane production completely stopped which required approximately 35 days. The digested microalgae samples were collected using a syringe twice a day and methane production was immediately measured. Data for each condition were collected both in a duplicate experiment and duplicate measurement. In addition to the methane yield, VS, soluble chemical oxygen demand (sCOD) and VFAs were also carefully recorded for each condition. Moreover, to clearly understand the methane yield and production rate, experimentally collected values of methane yields were modeled using a modified Gompertz equation as this model has an excellent capability to precisely forecast the lag phase which results mainly from the digestion of highly recalcitrant lignocellulosic biomass (Bohutskyi et al., 2018). The modified Gompertz equation is shown in Eq. (1):

$$G(t) = G_0 * \exp\left\{ -\exp\left[\frac{R_{max} * e}{G_0}(\lambda - t) + 1\right]\right\} \quad (1)$$

where, G(t) is methane yield at time t (mL/g-VS_{added}); G_0 is the ultimate methane yield (mL/g-VS_{added}); R_{max} is the maximal methane production rate (mL/g-VS_{added}.day); λ is the duration of lag phase (day); t is fermentation time (days); and e is the constant equivalent to Exp (1) = 2.718282.

To study the strength of linear association and correlation between different environmental factors, Pearson's correlation coefficient was employed. E-supplementary data of this work can be found in the online version of the paper. This correlation is the most commonly used method and is defined by Eq. (2)(Damtie and Choi, 2017):

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (2)$$

where r = correlation coefficient; x_i = values of the x-variables in the sample; $\bar{x}$ = mean of the values of x-variables; y_i= values of y-variables in the sample; and $\bar{y}$ = mean of the values of x-variables.

2.4. Analytical procedures and devices

Standard methods (APHA/AWWA/WEF, 2018) were employed to measure the TS, VS, total nitrogen (T-N), and total phosphorus (T-P). For analysis of soluble total phosphorus (sP), soluble total nitrogen (sN), ammonium (NH_4^+-N), and sCOD, samples were first filtered by using a 0.45 μm pore size filter and then analyzed by standard methods. tCOD measurement was conducted using the tCOD-chromium standard method following procedures described in ASTM D1252-95 (1995).

Carbohydrates were measured using the phenol–sulfuric acid method. This involves preparing 1 mL of sample in a test tube and adding 1 mL of double-distilled (dd) water, then mixing it with 0.05 mL 80% phenol. Then to each tube, 5.0 mL H_2SO_4 was rapidly added and mixed by a vortex test tube mixer. The samples were left to stand for 10 min and then placed in a 25 °C bath for another 10 min followed by an additional vortex. The final samples were then transferred to cuvettes and taken for reading the absorbance (at 470 nm spectra) against standard curve tubes which had been prepared using standard 100 mg glucose/L solution and dd water. Detailed procedures were taken from Nielsen (2010).

To determine protein concentrations, the bicinchoninic acid (BCA) method was employed using a Pierce™ BCA Protein Assay Kit from Thermo Scientific for application in the test tube procedure (Thermo Fisher Scientific Firm, 2009). The sample to working reagent (WR) ratio was 1:20. In this procedure, 0.1 mL of unknown sample replicates and standard samples were first mixed well with 2.0 mL of the WR in each tube. The tubes were then covered and incubated at 37 °C for 30 min and cooled. Then the absorbance of all the samples was measured within 10 min with the spectrophotometer set to 562 nm. The average 562 nm absorbance measurement of the blank standard replicates was subtracted from the 562 nm absorbance measurement of all other individual standard and unknown sample replicates. Then the protein concentrations of each unknown sample were read from the standard curve. Lipid concentrations were analyzed using gravimetry after extraction of lipids by solvents (chloroform: methanol, 1:2 *v/v*). For analyzing VFAs, a gas chromatograph (GC) (model YL6500 GC, Young Lin Instrument Co., Anyang-si, Korea) was used. To measure methane, total biogas volume was first determined and then multiplied by the percentage composition of methane (%) in the biogas which had been estimated from the gas chromatograph device stated above. To measure biogas volume, a water displacement method was employed after collecting the biogas samples in a two-liter, plastic Tedlar bag. In this biogas measuring technique, the volume of displaced water was assumed to be equal to the biogas volume. But corrections of biogas volumes to standard conditions have been applied to eliminate uncertainties induced due to information missing from different environmental factors such as temperature, pressure, water vapor and headspace composition. The method described in Strömberg et al. (2014) was used to calculate the adjustment factor even though the errors observed were not significant. OriginPro2016 compatible with Windows® 10 was employed to fit the data and estimate correlation coefficients (r^2) as well as to make graphs and

PCA analyses.

2.5. Microbial community analysis

Microbial communities were identified using the Illumina iSeq 100 sequencing system instructions (Illumina, CA, USA). The selected 16S rRNA gene region-specific sequences used as amplicon primers and the bacterial primers were 518F (5′-CCAGCAGCCGCGGTAATACG-3′) and 805R (5′- GACTACCAGGGTATCTAATCC-3′) targeting the V3-V4 region for bacteria (Yu et al., 2005). The DNA were PCR-amplified with the annealing temperature set to 58 °C and purified for construction of the library. The library was then paired-end (150 bp × 2) sequenced using the iSeq 100 platform. The paired reads were then merged and further processed by using short and low-quality sequences to reduce potential error. Operational taxonomic units were defined at a 97% sequence identity cutoff. According to the read, initially, a total of 65,368 to 94,706 sequence reads at the kingdom level were generated by sequencing 16S rRNA gene from 12 digester sludge samples. A total of 24 phyla, 47 classes, 100 orders, 221 families, and 588 genera were monitored from the total sequence reads. PCoA was also performed to visualize the similarity and dissimilarity of microbial community structures within and among the digesters based on the relative abundances of OTUs obtained from 12 different digester pretreatment samples. Moreover, a Circos graph showing the relationship between the pretreatment methods and their respective microbial abundance was constructed using Circos Table Viewer v0.63-9 © 2008-2020 (Krzywinski et al., 2009).

3. Results and discussion

3.1. Effect of pretreatment on microalgae hydrolysis

3.1.1. Release of organic macromolecules

Three pretreatment conditions (anaerobic pretreatment at 85 °C, anaerobic at 55 °C, and aerobic at 55 °C) at three different HRTs (3, 2, and 1 day) were selected to study the effects of biological pretreatment of microalgae on the release of macromolecules. As compared to the original concentrations of carbohydrate measured in the non-treated raw microalgae control sample (the diluted version of table 1), an average of 16-fold carbohydrate release was obtained from anaerobic pretreatment at 85 °C at an HRT of 3 days (Fig. 1a). For the 55 °C anaerobic and aerobic pretreatments, the average carbohydrates release was 13-fold and 7-fold, respectively. Similarly, the average protein release (Fig. 1b) at an HRT of 3 days at 85 °C anaerobic, 55 °C anaerobic, and 55 °C aerobic pretreatments increased by 10-fold, 6-fold, and 3-fold, respectively, compared to their initial control values. These trials showed that 85 °C was relatively the better temperature and anaerobic was the better redox condition, and longer HRTs such as 3 days were the optimum retention times for the highest releases of carbohydrates and proteins. Such variations in performance might be attributable to differences in their microbial activity because at 85 °C there was release only of soluble macromolecules with no further biodegradation; on the other hand, at 55 °C the relative amount of carbohydrate concentration was quite low due to an active thermophilic biological process which further consumed and/or decomposed the carbohydrates into intermediate products such as VFAs (Solé-Bundó et al., 2017). Under aerobic conditions, there might also be high microbial activity consuming soluble organics and producing CO_2, which may explain why the lowest concentration of such macromolecules occurred in aerobic conditions. E-supplementary data regarding VS removal can be found in the online version of this paper. On the other hand, the highest release of lipids was obtained at 55 °C during aerobic pretreatment at an HRT of 3 days (Fig. 1c), showing an average of 2-fold increment as compared to its initial lipids concentration in the control emphasizing that lipids release was linked mainly to the degree of microbial activity which can release both intracellular and extracellular materials rather than to the thermal effect which mainly avails lipids from the extracellular media (Solé-Bundó et al., 2017).

3.1.2. Volatile fatty acids

The VFAs generated during AD in this study included acetic acid (C2), propionic acid (C3), butyric acid (C4), and valeric acid (C5). As per results from the anaerobic pretreatment at 85 °C (Fig. 2a), a substantial amount of long-chain VFAs such as valeric and isovaleric acids were produced, whereas at 55 °C (both anaerobic and aerobic), the composition was dominated by short-chain VFAs such as acetic, butyric, and propionic acids (see Fig. 2b). Furthermore, concentrations of the VFAs more than doubled in 55 °C anaerobic pretreatments at an HRT of 1 day (5.3 g/L) as compared to 85 °C anaerobic at an HRT of 1 day (2.3 g/L). These differences might be ascribed to microbial hydrolysis activity in that at hyper-thermophilic temperatures, solubilization could be the result of physicochemical factors, while at thermophilic temperatures, there are biological factors from the action of microbial enzymes. Temperature is generally a vital parameter in acidogenesis due to its direct contribution both to microbial growth and system metabolism (Wainaina et al., 2019).

In aerobic pretreatment, the quantity and structure of the VFAs spectrum significantly changed. In anaerobic pretreatment, released organic substances from the disruption of cell walls continue to accumulate as VFAs, and any further retention of the substrate yields even relatively more short-chain VFAs. On the other hand, in the aerobic process, once soluble organics and short-chain VFAs are formed, the microbes tend to consume and form CO_2 and greatly reduce the concentration of short-chain VFAs. This is one reason why the magnitude of total VFAs in the aerobic digester was also significantly less than under anaerobic conditions (Fig. 2c,d). The second reason for considerably higher VFAs formation in anaerobic than in aerobic pretreatment might be linked to microbial diversity and abundance. It is generally recognized that the phyla *Firmicutes and* Thermotogae are highly correlated with VFAs formation (Lin et al., 2017). Microbes detected in aerobic pretreatments are mainly heterotrophic bacteria with tendencies that stimulate the digestion of slowly biodegradable compounds through exoenzymes, resulting in highly destructed VS and VFAs being converted to CO_2 (See detailed microbial discussions in Section 3.3). The most dominant VFAs in anaerobic pretreatment were acetic and propionic acids which formed a high concentration of short-chain VFAs (Fig. 2d). Under aerobic conditions, butyric acid was the predominant VFA (Fig. 2c) because the system initially accumulates a substantial amount of acetic acid for ATP production but this excess accumulation might induce microbes to activate reverse fermentation which ultimately converts acetic acid to butyric and propionic acids (Fig. 1c) (Jin et al., 2016). In addition, high concentrations of butyric acid and other long-chain VFAs under aerobic conditions might be attributable to the high concentrations of lipids released during the aerobic pretreatment from disruption of algae that changes lipid molecules to long-chain fatty acids by microbial action.

The effect of HRTs on VFAs formation was also studied. At 85 °C, there was relatively more concentration of short-chain VFAs at an HRT of 3 days (Fig. 2a) because longer HRTs in hyper-thermophilic working temperatures might lead to an increase in the rate of hydrolysis, making hydroxylates theoretically available for fermentative microbial metabolism (Wainaina et al., 2019). Moreover, maintaining the substrate at high temperatures for an extended period favors further degradation of the long-chain VFAs.

3.1.3. Biomethanization

As can be seen from the experimental results in Fig. 3, each pretreatment technique offers its unique features with regards to biomethanization. E-supplementary data of the modeling work can be found in the online version of the paper. Anaerobic pretreatment at 85 °C showed the highest methane yield (G_o = 239.93 mL/g-VS_{added}); anaerobic pretreatment at 55 °C had the highest rate of methane yield

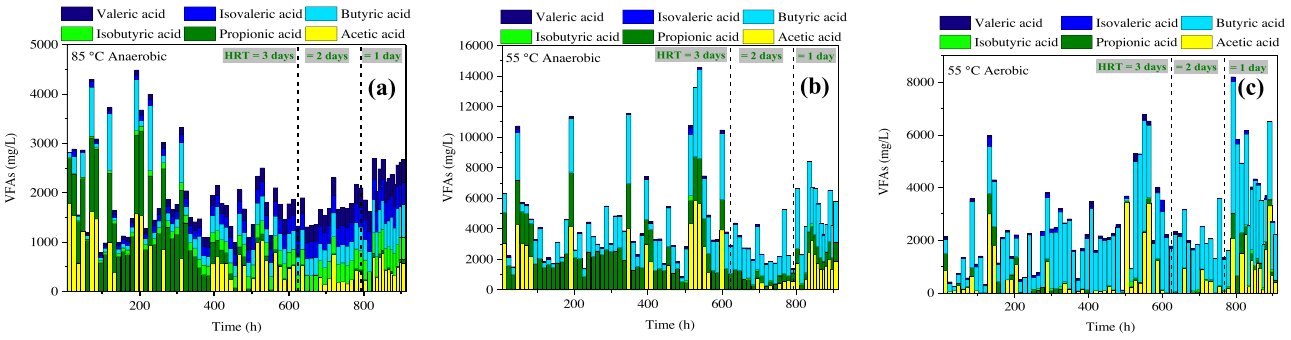


Fig. 2. Amounts and types of VFAs during different types of biological pretreatments from (a) anaerobic pretreatment at 85 °C; (b) anaerobic pretreatment at 55 °C; (c) aerobic pretreatment at 55 °C.

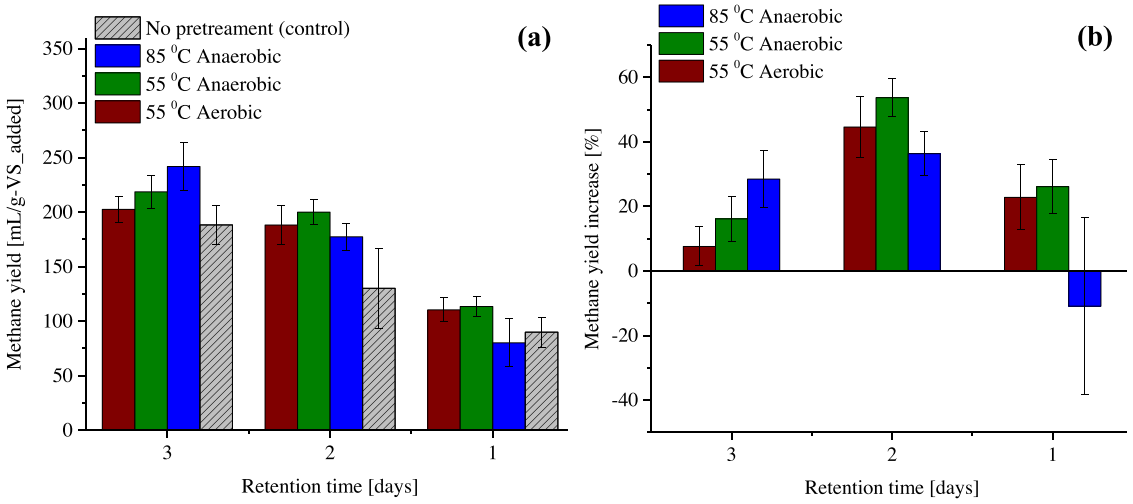


Fig. 3. Experimentally measured methane yields of microalgae pretreated using different pretreatment conditions: (a) absolute methane yield; (b) relative percentage of methane yield increase with respect to the control.

(R_{max} = 30.7 mL/g-VS_{added}/day); and aerobic pretreatment at 55 °C resulted in the smallest lag-time (λ = 0.06 days). Because of the very high rate of methane production at anaerobic pretreatment at 55 °C, one would speculate a priori that the highest methane yield would have been observed at 55 °C. However, the decline at an HRT of 3 days might be due to the long HRT employed in which methane production had already been partially started and squandered during the pretreatment stage. On the other hand, the methane yield reduction at an HRT of 3 days after aerobic pretreatment might reflect the sudden change in redox or microbial composition (Section 3.3) or the loss of the solubilized organics by biodegradation. Another reason for reduction of methane in aerobic pretreatment may be due to the inhibition of methane production caused by the formation of intermediate products such as long-chain VFAs from lipid molecules (Ward et al., 2014). The results showed that the methane yields generally appeared to correlate strongly with the HRT and each pretreatment had its own optimal HRT. Those with very rapid biological activities required shorter HRTs. In our study, at an HRT of 2 days, a maximum methane yield enhancement up to 45%, 54%, and 36% was observed from aerobic pretreatment at 55 °C, anaerobic pretreatment at 55 °C, and anaerobic at 85 °C, respectively, as compared to the samples without any pretreatment. Similar methane enhancement (nearly 55%) has also been observed from primary sludge digestion (Solé-Bundó et al., 2019).

3.2. Effect of pretreatment on the interrelationship between environmental factors

Two PCAs accounting for approximately 68% of total variance were considered (Fig. 4). The first principal component (PC1) expressed 43.1% of the overall variance. Parameters linked to PC1 were total VFAs, acetic acid, lipids, removed VS, and butyric acid which are positively correlated with each other but negatively affected by proteins, carbohydrates, valeric acid, isovaleric acid, and isobutyric acid, all of which overlapped and correlated well with each other. PC2 represented 24.9% of variability and showed a strong correlation with factors such as propionic acid, sN, sP, methane yield, ammonia, and sCOD which also correlated with each other. Loading all the pretreatment methods on the factorial plane described by PC1 versus PC2 formed different clusters. The distance between each point describes the strength/weakness of correlation and also emphasizes the importance of the redox condition, HRT, and temperature. Based on the dominant background process, the clusters were named as a hyper-thermo-anaerobic cluster (green dots of AnH1, AnH2, and AnH3), thermo-anaerobic cluster (black dots of AnT2 and AnT3), and thermo-aerobic cluster (red dots of AT1, AT2, AT3, and AnT1). Results obtained from an HRT of day 1, especially in the thermo-clusters, were erroneous and were excluded from the discussion because of their uncertainty. Normally, too short a HRT may cause incomplete substrate degradation or microbial death by starvation and may not represent reliable data (Magdalena et al., 2019).

The hyper-thermo-anaerobic cluster can best be explained by carbohydrates, proteins, and release of relatively long-chain VFAs (valeric,

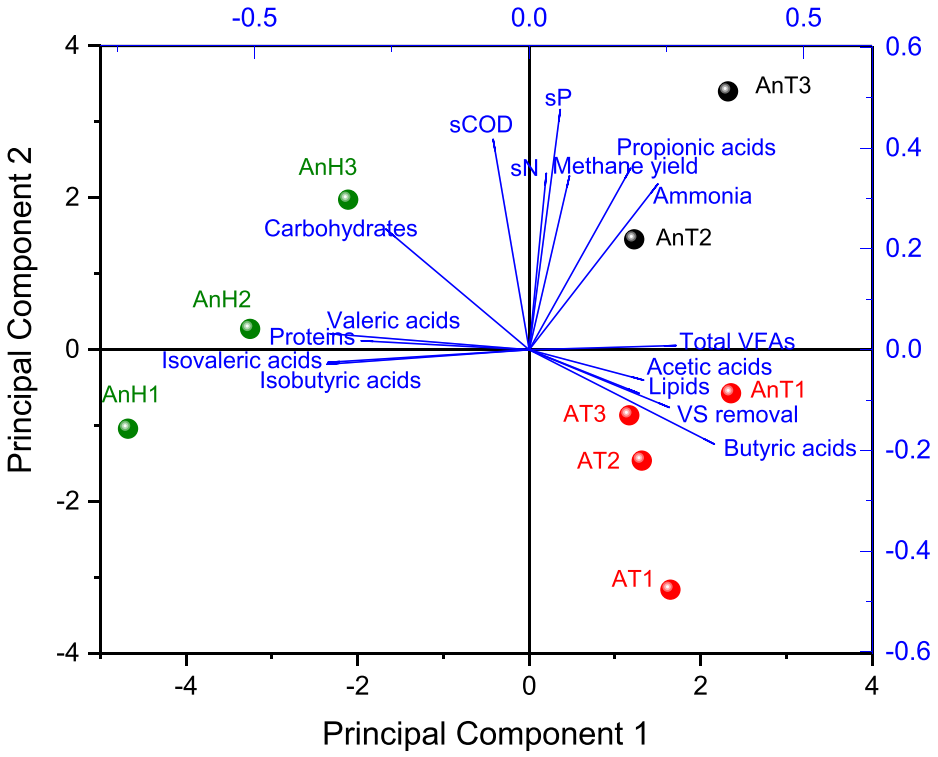


Fig. 4. Principal component analysis to study the relationships between different pretreatment methods (colored dots) and environmental factors (blue arrows). *Note:* In the abbreviation AT1, AT2, AT3, AnT1, AnT2, AnT3, AnH1, AnH2, AnH3, "H" refers to hyperthermophilic temperature (85 °C); "T" represent thermophilic temperature (55 °C); "An" represent anaerobic pretreatment process; and "A" denotes aerobic pretreatment process. The numbers 1, 2, and 3 stand for the HRTs (days).

isovaleric, and isobutyric acids) which also correlate well with each other. This confirms the findings of Passos et al. (2013) that pretreatment at high temperature (85 °C) succeeded in solubilizing compounds that remained intact when a lower temperature (55 °C) was applied. Normally, microalgae are not likely to disintegrate at 70–90 °C but temperatures in this range enhance methane yield by partial weakening and breaking down cell walls and releasing macromolecules (Passos et al., 2014). The slow soluble organic consumption induced by weak microbial activity also favors the accumulation of these macromolecules. An interesting observation is that when long-chain VFAs (isovaleric, iso-butyric, and valeric acids) were produced, total VFAs were reduced. This is because the formation of long-chain VFAs in the reactor was a clear indication of the presence of some internal/external inhibiting factors that slow down the entire biodegradation process which, in turn, affected total VFAs as well. In addition, the long-chain VFAs themselves can also inhibit acetogenesis, thereby decreasing acetic acid and leading to a decrease in total VFAs. This interpretation could be supported by the current PCA result (Fig. 4) in which acetic acid had a strong correlation with total VFAs.

The thermo-aerobic cluster was best explained by butyric acid, lipids, acetic acid, total VFAs, and VS removal implying that aerobic conditions are a relatively good choice if the lipids and total VFAs were estimated. The highest accumulation of lipids in the aerobic pretreatment might be attributable to a lack of lipid digesting microbes (e,g *Bacteroidetes*). The high VS destruction was due to improved solubility of the organic matter by aerobic conditions that rapidly avail biodegradable organic matter to the oxidizing biomass (Montalvo et al., 2016) such as *Symbiobacterium*, *Soehngenia,* and *Ureibacillus* (Section 3.3). Due to high reduction of VS under thermo-aerobic conditions, such a pretreatment protocol is generally not an appropriate technology for methane production as the microbes convert the soluble organics to CO_2 gas; otherwise, relatively short HRTs need to be employed for the pretreatment. This is also the reason for the lack of correlation observed between removal of VS and methane yield in the thermo-anaerobic cluster. On the other hand, aerobic pretreatment is advantageous, being a powerful rapid process with the pH and temperature environment conducive to biomethanization (Jang et al., 2014).

Even though the highest concentration of total VFAs was recorded under thermo-anaerobic conditions, no correlation was observed between the thermo-anaerobic cluster and total VFAs in the PCA analysis. This reflects the total VFAs being highly dominated by the butyric acid concentration which had been abundantly formed in the thermo-aerobic pretreatment. The strong correlation between the thermo-aerobic cluster and butyric acid may be due to higher hydrogen partial pressure formation in the system. Usually, butyric acid accumulates in the system when hydrogen-utilizing microbes are exposed to partial pressures above 10^{-4} atm (Magdalena et al., 2019). In our experiment, excess butyric acid thus formed a strong correlation between the thermo-aerobic pretreatment and total VFAs because over 71% of the total VFAs originated from the butyric acid. The remainder of the total VFAs was metabolized mainly to acetic acid (21%) which also correlates well with the thermos-aerobic pretreatment. In aerated reactors like AT2 and AT3, syntrophic propionate oxidation (SPO) is highly favored and microaeration enhanced the oxidation of propionate to acetate and stimulated the formation of high acetic acid concentration (Lim and Wang, 2013) even though part of it was later converted to butyric acid during the reverse fermentation process. This butyric-acetic interaction and accumulation was responsible for the strong correlation between acetic and butyric acids in the thermo-aerobic cluster. On the other hand, propionic acid formation was correlated more with the thermo-anaerobic cluster than the thermo-aerobic cluster. This is mainly because the free ammonia nitrogen accumulation in AnT2 and AnT3 was less than the inhibition level; thus, as compared to syntrophic acetate oxidation and hydrogenotrophic methanogenesis, acetoclastic methanogenesis becomes the thermodynamically predominant methane pathway and leads to the accumulation of acetic acid, hydrogen, and formate molecules in the reactors. Consequently, SPO becomes less thermodynamically favorable and jeopardizes the growth of sensitive and slow-growing syntrophic propionate oxidizers (Capson-Tojo et al., 2017) that leads to the formation and accumulation of propionic acid. The strong correlation between propionic acid with a thermo-anaerobic cluster and butyric with a thermo-aerobic cluster was attributed to the

total VFA spectra of each cluster which showed ratios of acetic: propionic: butyric acid of 20.1:25.4:51.5 and 20.1:4.6:71.5, respectively. High concentrations of short-chain VFAs were measured during the thermo-anaerobic pretreatments and are the main reason for the high correlation between methane yield and propionic acid. Moreover, acetate is considered to be a direct substrate for methanogens, and the reduced levels of propionate and increased production of acetate after pretreatment accelerated the subsequent methanogenesis as reported by Lim and Wang (2013). Methane yield lacks correlation with total VFA because its spectrum in both the aerobic and anaerobic pretreatments was dominated by butyric and propionic acids which are not a direct substrate for the methanogens. On the other hand, a direct correlation was observed between methane yields with nutrients (sN, sP) and ammonia because cell lysis and disruption of complex cell wall structures which involved protein fermentation, dephosphorylation, deamination, and methanogenesis released these highly correlated end products in a similar trend (Ayala-Parra et al., 2017).

3.3. Effect of pretreatment on microbial diversity and abundance

In 85 °C anaerobic reactors, *Firmicutes and Proteobacteria* were the most dominant phyla accounting for over 98% of total bacterial community (Fig. 5a). A limited amount of *Yersinia and Lactobacillus* were observed at the genus level in all HRTs except for high amounts of *Clostridium* at an HRT of 3 days (Fig. 5c). However, at such a high temperature, it is unlikely that thermophilic microbes would exist except for their DNA (Damtie et al., 2020). Therefore, no bacterial activity and biodegradation would take place and the pretreatment actions from these conditions enhance bioavailability from their thermal effects and result in the release of undigested organic macromolecules (carbohydrates and proteins).

In anaerobic 55 °C conditions, hydrolytic fermentative phyla such as *Thermotogae, Bacteroidetes, Chloroflexi, Cyanobacteria,* and *Actinobacteria* were observed to be dominant at all HRTs. *Firmicutes* play mainly a major role in the entire biodegradation activities; Chloroflexi and *Thermotogae* have the metabolic ability to digest carbohydrates while Proteobacteria and Bacteriodetes decompose proteins and amino acids in syntrophic association with hydrogenotrophic methanogens (Gonzalez-Fernandez et al., 2015). High VFAs formation in anaerobic 55 °C conditions was also linked to *Thermotogae* and *Bacteroidetes*. Moreover, at the genus level, *Clostridium, Sporanaerobacter, Thermogemmatispora, Enterococcus, Calothrix, Blautia, and Ruminococcus* were detected. Among these, *Clostridium* was found in the highest concentration (over 46%) in all HRTs and may be most responsible for the superior biological activity and the high tendency of hydrolysis in anaerobic pretreatment at 55 °C. *Clostridia* are chemoorganotrophic and saccharolytic fermentative microbes mainly responsible for the hydrolysis of macromolecules, syntrophic acetate oxidation, and hydrogenotrophic methanogenesis. This is principally because they have cellulosomes, which are extracellular complex enzyme systems containing various enzymes such as endoglucanases, exoglucanases, hemicellulases, chitinases, pectin lyases, and lichenases that can easily degrade lignocellulosic materials to soluble monomers (Salakkam et al., 2019). *Clostridia* are also involved in butyrate, acetic acid, lactic acid, and ethanol production due to their ability to carry out mixed acid and alcohol fermentation (Magdalena et al., 2019). They also contain species such as C. thermocellum that are highly thermophilic and cellulolytic and whose spores can survive under extreme conditions (Lin et al., 2017). Therefore, these microbes can be responsible for the superior biodegradation of carbohydrate macromolecules that resulted in either directly forming short-chain VFAs (Fig. 2) or indirectly supporting methane yield (Fig. 3a). The high biodegradation rate by Clostridia can also be explained using the sCOD values. Moreover, high ammonia release in the anaerobic pretreatment at 55 °C may also be attributable to the proteolytic action of *Clostridia* which accelerated the hydrolysis of proteins into amino acids. E-supplementary data of sCOD values and ammonia release can be found in the online version of the paper. These amino acids are then further fermented to obtain energy via the Stickland reaction that involves coupled oxidation and reduction of amino acids to form VFAs and ammonia (Magdalena et al., 2018). *Thermogemmatispora* and Sporanaerobacter were the next most dominant at the genus level. *Thermogemmatispora* is a cellulolytic and sporulating thermophile having excellent nitrate reduction and polysaccharides degradation at high-temperatures (Yabe et al., 2017). Sporanaerobacter is a thermophile anaerobe that utilizes some sugars, peptides, and various single amino acids to form acetic acid (Klassen et al., 2017). Results showed that the species observed at HRTs of 1, 2, and 3 days both at the phylum and genus level were quite similar (Fig. 5b). This emphasizes the effect of HRT in different pretreatment methods.

In aerobic 55 °C conditions, *Firmicutes* and *Proteobacteria* were the only species responsible for the prevailing biological activities. At the genus level, *Luteimonas, Symbiobacterium, Soehngenia, Thermobacillus, Chromobacterium,* and *Ureibacillus* were dominant. Unlike anaerobic pretreatment, significant changes in the microbial community were observed with varying HRTs (Fig. 5c green dots) which highlight a strong correlation between HRT and microbial diversity in aerobic pretreatment.

At an HRT of 2 days, *Ureibacillus* and *Soehngenia* were dominant. Ureibacillus (34%) are thermophilic aerobic endospore-forming bacteria capable of high removal of COD and solids by releasing lytic enzymes. Their actions include nitrate reduction, starch hydrolysis, and acidogenesis from polysaccharides (Fortina et al., 2001). They also provide excellent routes for enhancing soluble organic matters (sCOD) at shorter HRTs for bioavailability and rapid degradation of organic substrates at 55 °C at longer HRTs (Liu et al., 2010). The *thermosphaericus* species, Ureibacillus, interestingly, also doesn't consume sugars but decomposes lignocelluloses molecules (Okuda et al., 2008). On the other hand, *Soehngenia* (23%), a carbohydrate degrader, is a mesophilic aerotolerant anaerobic bacteria, having saccharolytic and weakly proteolytic behavior which also fix molecular nitrogen. Major byproducts of fermentation include formate, H_2, CO_2, acetate, and ethanol (Parshina et al., 2003). A marked increase in methane yield obtained at an HRT of 2 days (Fig. 3b) perhaps resulted from the initiation of these microbes' hydrolysis of macromolecules.

At an HRT of 3 days, *Luteimonas, Symbiobacterium, Thermobacillus, Ureibacillus* were dominant. *Luteimonas* (23%) are aerobic microbes able to reduce nitrate to nitrite while having a negative effect on glucose fermentation (Sun et al., 2012). They degrade organic matter degradation from lignocellulose-rich digestates only at mesophilic conditions (Akyol et al., 2019). This genus may make only limited contributions as it may not exist at 55 °C and has less tendency to facilitate fermentation. However, *Symbiobacterium* (22%) was probably responsible for syntrophic relationships with supporting bacteria such as *thermophilic Bacillus* sp. which oxidizes acetate or ethanol to produce energy (Dessì et al., 2019). Therefore, its significant relative abundance in the aerobic reactor can be linked to the prevalence of *Ureibacillus* (12%) and *Thermobacillus* (18%) microbes that have substantial ability to degrade. *Symbiobacterium* has been reported to consume various forms of organic matter, such as carbohydrates and amino acids and produce L-tyrosine, phenol-lyase, and thermostable enzymes responsible for the removal of large amounts of sCOD and tCOD as well as facilitating degradation of VFAs owing to its tendency to disinhibit acetate degradation (Jin et al., 2016). The highest VS reduction observed at an HRT of 3 days might be attributable to these genera: *Symbiobacterium, Thermobacillus* and *Ureibacillus*. Moreover, a substantial amount of *Thermobacillus and Clostridium* has also been detected in both HRTs. *Thermobacillus* are aerobic thermophiles (up to 72 °C) possessing the greatest potential to breakdown lignocellulose and are known as plant biomass degraders. These could biodegrade cellulose and polysaccharides to derive energy for microbial activities using complex cellulolytic and hemicellulolytic enzymes (Lemos et al., 2017). Detection of *Clostridium* (up to 5% concentration) and some other anaerobes indicates the thermophilic aerobic

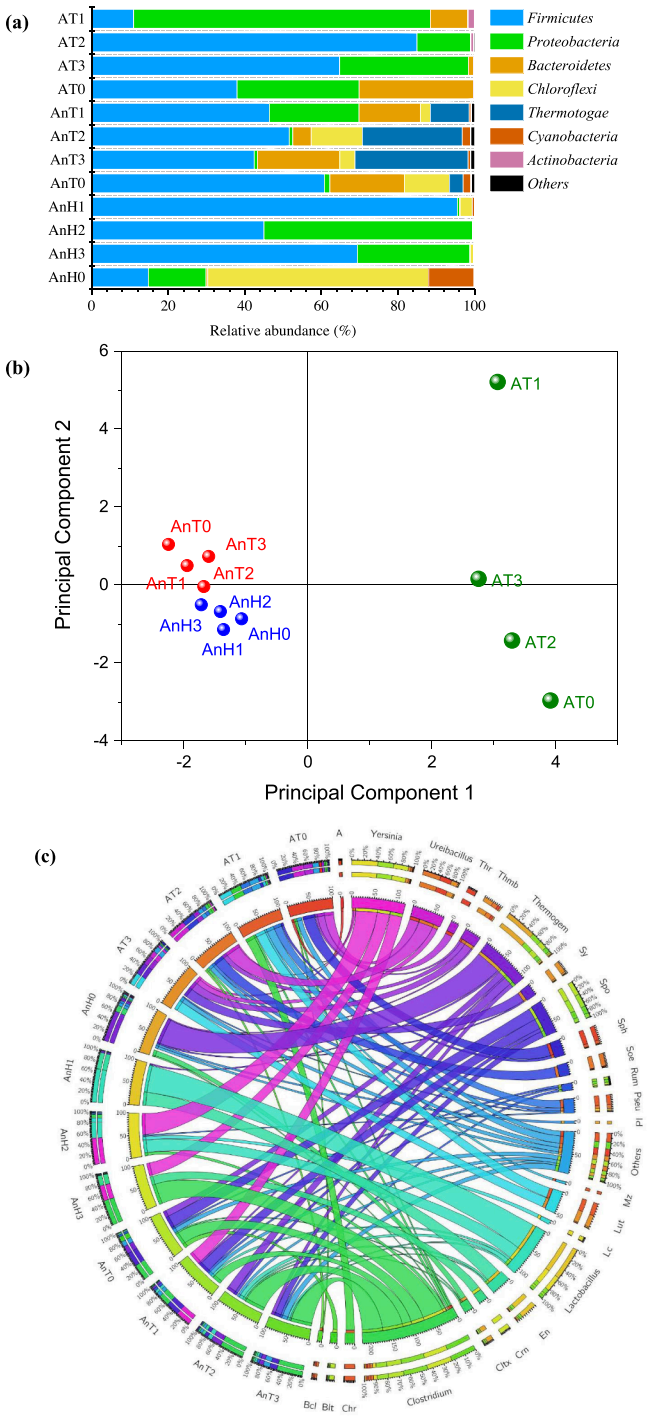


Fig. 5. (a) The relative abundance of microbial species at the phylum level (b) Principal component analysis for clustering different pretreatment methods based on genus-level diversity data (c) The interrelationship network and functional category between various microalgae pretreatment methods and respective microbial species at the genus level based on their relative abundance. Major species include Clostridium, Yersinia, Lactobacillus, Ureibacillus, Thermogemmatispora (Thermogem), Sporanaerobacter (Spo), Luteimonas (Lut), Enterococcus (En) Thermobacillus (Thmb), Sphingobacterium (Sph), Pseudomonas (Pseu), Soehngenia (Soe), ymbiobacterium (Sy), Chromobacterium (Chr), Carnobacterium (Crn), Thauera (Thr), Calothrix (Cltx), Blautia (Blt), Ruminococcus (Rum), Bacillus (Bcl), Lactococcus (Lc), Mesorhizobium (Mz), Acinetobacter (Ac), Plesiomonas (Pl). *Note:* Only genera with coverage higher than 3% and phyla coverage higher than 1% of the total number of sequences of each sample have been considered. The rest were categorized as "others". Involved pretreatment methods were anaerobic digestion at a temperature of 55 °C [AnT0, AnT1, AnT2, AnT3] and at a temperature of 85 ^{0}C [AnH0, AnH1, AnH2, AnH3] as well as aerobic pretreatment at a temperature of 55 ^{0}C [AT0, AT1, AT2, AT3]. AnH0, AT0, and AnT0 correspond to the initial condition before starting the data collection." T" represents thermophile and "H" represents hyperthermophile. HRTs (days) are represented by 1,2,and 3.

process in the current study was not entirely aerobic due to limitations in dissolving and transferring oxygen at high temperatures and dense substrate concentrations.

4. Conclusions

sN, sP, sCOD, and propionic acid concentration can serve as reliable proxies to optimize and predict the BMP of microalgae biodegradation. Total VFAs formation was highly correlated with acetic, butyric acids, and removal of VS. Hydrolysis and biomethanization were not correlated during microalgae biodegradation in TPAD. Anaerobic pretreatments at 85 °C showed excellent performance in releasing carbohydrates and proteins, but anaerobic pretreatments at 55 °C demonstrated high total VFAs due to dominancy of *Clostridium*. The predominance of *Luteimonas*, *Symbiobacterium, Soehngenia, Thermobacillus,* and *Ureibacillus* genera was responsible for VS removal and formation of long-chain VFAs in aerobic pretreatments at 55 °C.

CRediT authorship contribution statement

Mekdimu Mezemir Damtie: Conceptualization, Investigation, Data curation, Visualization, Writing - original draft, Writing - review & editing. **Jingyeong Shin:** Conceptualization, Methodology, Investigation, Validation, Formal analysis, Writing - original draft, Writing - review & editing. **Hyun Min Jang:** Formal analysis, Writing - review & editing. **Hyun Uk Cho:** Formal analysis, Investigation, Writing - review & editing. **Jinhua Wang:** Formal analysis, Writing - review & editing. **Young Mo Kim:** Conceptualization, Writing - review & editing, Supervision, Resources, Funding acquisition, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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